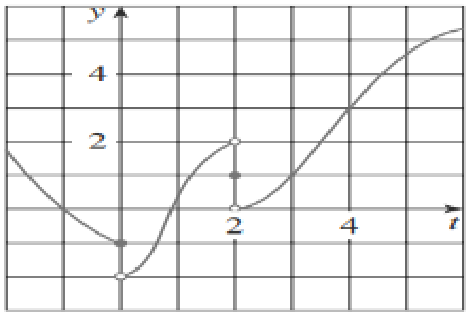
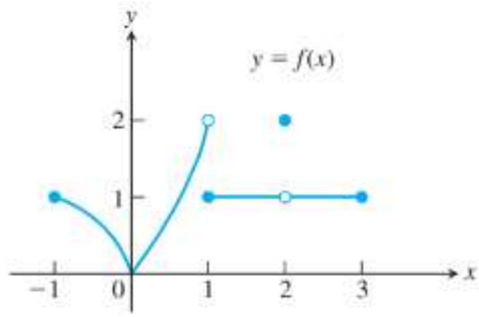
Assignment-2

Q.1 Form the graph of the function g(t), state the value, limit, and continuity at each point, if exists. If it does not exist, explain why? Also discuss vertical and horizontal asymptotes of g(t).

(a) $\lim\limits_{x\to 0^+} g(t)$ (b) $\lim\limits_{x\to 0^-} g(t)$ (c) $\lim\limits_{x\to 0} g(t)$ (d) $\lim\limits_{x\to 2^+} g(t)$

(e) $\lim\limits_{x\to 2^-} g(t)$ (f) $\lim\limits_{x\to 2} g(t)$ (g) $\lim\limits_{x\to 4^+} g(t)$ (h) $\lim\limits_{x\to -1} g(t)$





Q.2 Find limits of the given functions

I. $\lim\limits_{t\to 0} \dfrac{\sqrt{t^2+9}-3}{t^2}$

II. $\lim\limits_{h\to 0} \dfrac{(4+h)^2-16}{h}$

III. $\lim\limits_{x\to 0} \dfrac{\sin^2 x + 4\cos 3x}{2\tan^2 x}$

IV. $\lim\limits_{x\to \pi/3} \dfrac{\arctan(x^3-x)}{2\sec^2 x}$

V. $\lim\limits_{t\to 0} \left(\dfrac{1}{t\sqrt{1+t}} - \dfrac{1}{t}\right)$

VI. $\lim\limits_{x\to 1} \dfrac{\sqrt{x}-x^2}{1-\sqrt{x}}$

VII. $\lim\limits_{x\to 1} \dfrac{\sqrt[3]{x}-1}{\sqrt{x}-1}$

VIII. $\lim\limits_{x\to 0} \dfrac{1+|1-x|}{x}$

IX. $\lim\limits_{h\to 0} \dfrac{h^2|h+2|}{|h^2-2h+1|}$

Q.3 Find the types of discontiniouty of the functionsat given point

(a) $f(x)=\begin{cases}1+x^2 & ; x<1\\ 4-x & ; x\geq 1\end{cases}$

(b) $f(x)=\begin{cases}e^x & ; x<0\\ x^2 & ; x\geq 0\end{cases}$

(c) $f(x)=\begin{cases}\dfrac{x^2-x}{x^2-1} & ; x\neq 1\\ 1 & ; x=1\end{cases}$

(d) $\lim\limits_{x\to\pi/2}\dfrac{e^{\tan x}}{\cos^2 x}$

Q.4 Find constants for which given functions are continuous everywhere

(a) $f(x)=\begin{cases}x^2-c^2 & ; x<4\\ cx+20 & ; x\geq 4\end{cases}$

(b) $\lim\limits_{x\to 0}\dfrac{\sqrt[3]{1+ax}-1}{x}$

(c) $f(x)=\begin{cases}n & ; x=\pi/4\\ m\tan^2 x+1 & ; x<\pi/4\\ \sin^3 2x & ; x>\pi/4\end{cases}$

d) $f(x)=\begin{cases}ax+b & ; x\leq 0\\ x^2+3x-b & ; 0<x\leq 2\\ 3x-5 & ; x>2\end{cases}$

Q.5 Find the limit at infinity and horizontal asymptotes for the following functions

a) $y=\dfrac{1-x-x^2}{2x^2-7}$

b) $y=\mathrm{Tan}^{-1}\left(x^4-x^2\right)$

c) $y=\cos\left(\sqrt{x}\right)$

d) $y=\dfrac{t^2+2}{t^3+t^2-1}$

e) $y=\dfrac{x+2}{\sqrt{9x^2+1}}$

f) $y=\dfrac{\sqrt{9x^6-x}}{x^3+1}$

g) $y=\dfrac{\cos^2 x.\csc x}{1+\tan^2 x}$

h) $y=\left(\sqrt{9x^6+x}-3x\right)$

i) $y=\left(\sqrt{x^2+2x}+x\right)$

Q.6 Find the vertical asymptotes of the given curves

a) $y=\dfrac{x}{x+4}$

b) $y=\dfrac{x^2+4}{x^2-1}$

c) $y=\dfrac{x^3}{x^2+3x-10}$

d) $y=\dfrac{x^3+1}{x^2+x}$

Q.7 Find the interval of continuity and point of discontinuity for the following functions

a) $y=e^{\mathrm{Tan}^{-1}x}$

b) $y=\dfrac{x^2}{1+\ln x}$

c) $y=\dfrac{1+e^{2x}}{1-e^x}$

d) $y=\dfrac{x^2+2x+5}{(1-x)(2+x)(3-x)}$